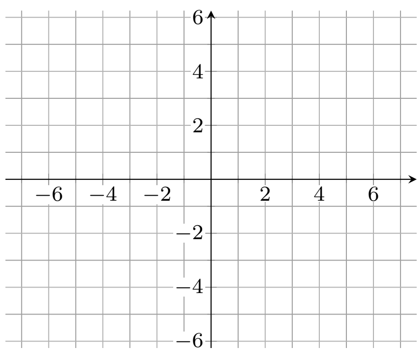
1.6: Absolute Value Functions

Definition. (Absolute Value Function)
The absolute value function can be defined as a piecewise function

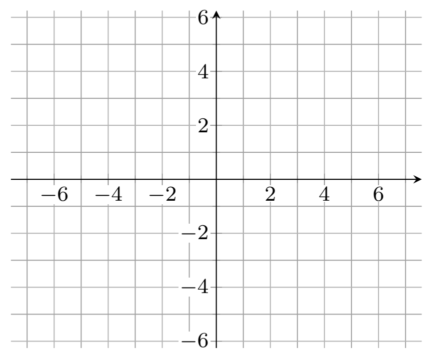
$$f(x) = |x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

Example. Graph the following

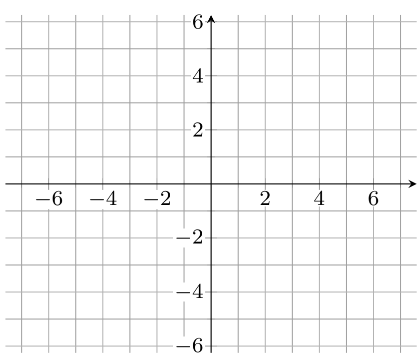
$f(x) = |x|$



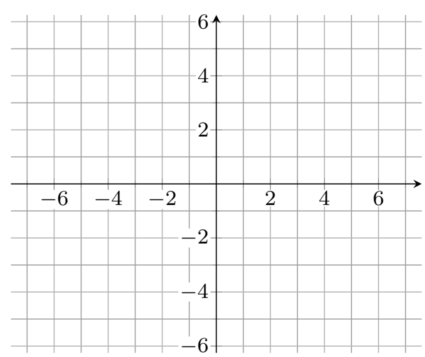
$f(x) = |x - 2|$



$f(x) = |x| + 3$



$f(x) = 2|x - 3| + 1$



Example. Solve the following equations Graphs

$|x| = 3$

$|2x + 6| = 2$

$|x + 3| - 1 = 0$

$2 - |4 - x| = 0$

Example. Solve the following inequalities. Give your answer in interval notation. Graphs

$|x-5| \leq 4$

$\left|\frac{x}{4}-1\right| < 2$

$|2-x| \leq 3$

$|x-3| > 2$